D 30562 **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Mathematics

MTS 5B 08—LINEAR PROGRAMMING

(2019 Admissions only)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions. Each question carries 2 *marks.*
Ceiling is 20.

1. Convert the following linear programming problem into canonical form :

   Maximize $f(x, y) = -2y - x$ subject to

   $$2x - y \geq -1$$
   $$3y - x \leq 8$$
   $$x, y \geq 0.$$

2. Draw and shade the region of a bounded polyhedral convex subset of $\mathbb{R}^2$ in the first quadrant.

3. Represent the following linear programming problem into canonical slack form :

   Maximize $f(x_1, x_2) = 200x_1 + 150x_2$ subject to

   $$x_1 + 2x_2 \leq 20$$
   $$2x_1 + 2x_2 \leq 30$$
   $$2x_1 + x_2 \leq 25$$
   $$x_1, x_2 \geq 0.$$

4. Define infeasible linear programming problem.

5. Pivot on 4 in the canonical maximum tableau given below :

x	y	-1	
1	2	3	$= -t_1$
4	5	6	$= -t_2$
7	8	9	$= f$

Turn over

6. Write the simplex algorithm for minimum tableaus.

7. What do you mean by complementary slackness ?

8. State Duality theorem.

9. Consider the canonical minimization linear programming problem given below :

Minimize $g(y_1, y_2) = -y_2$ subject to

$$y_1 - y_2 \geq 1$$
$$-y_1 + y_2 \geq 2$$
$$y_1, y_2 \geq 0.$$

State the dual canonical maximization of the linear programming problem.

10. Using VAM to obtain an initial basic feasible solution of the transportation problem given below :

3	2	1	30
2	5	9	75
40	30	50	

11. Write the Hungarian algorithm for solving assignment problem.

12. Explain Northwest-corner method for obtaining initial basic feasible solution in transportation problem.

Section B

Answer any number of questions.
Each question carries 5 *marks.*
Ceiling is 30.

13. Solve the following linear programming problem by geometrical method :

Maximize $f(x, y) = 5x + 2y$ subject to

$$x + 3y \leq 14$$
$$2x + y \leq 8$$
$$x, y \geq 0.$$

14. Solve the following canonical linear programming problem using simplex algorithm :

x_1	x_2	-1	
-1	1	1	$= -t_1$
1	-1	3	$= -t_2$
1	2	0	$= f$

15. Solve the following canonical linear programming problem by simplex method :

x_1	1	-1	-2
x_2	1	-1	4
-1	3	-2	0
	$= t_1$	$= t_2$	g

16. Solve the non-canonical linear programming problem given below :

Maximize $f(x, y) = x + 3y$ subject to

$$x + 2y \leq 10$$
$$3x + y \leq 15$$
$$x \geq 0.$$

17. Write the dual simplex algorithm for minimum tableaus.

18. A company wishes to assign five of its worker to five different jobs (one worker to each job and vice versa). The rating of each with respect to each job on a scale of 0 to 10 (10 being high rating) is given by the following table :

	J_1	J_2	J_3	J_4	J_5
W_1	5	4	2	8	5
W_2	7	6	4	6	9
W_3	5	5	3	3	2
W_4	4	3	5	5	4
W_5	3	6	4	10	2

If the company wishes to maximize the total rating of the assignment, find the optimal assignment plan and corresponding maximum total rating.

19. Apply the minimum entry method to obtain the initial basic feasible solution for the transportation problem given below :

4	5	5
3	2	7
6	3	9
7	5	4
14	11	

Turn over

Section C

Answer any **one** *question.*
Each question carries 10 *marks.*
Maximum marks 10.

20. Apply simplex algorithm to the maximum tableau :

x_1	x_2	-1	
-1	-2	-3	$= -t_1$
1	1	3	$= -t_2$
1	1	2	$= -t_3$
-2	4	0	$= f$

21. Write transportation algorithm. Using this algorithm solve the following transportation problem :

5	12	8	50	26
11	4	10	8	20
14	50	1	9	30
15	20	26	15	

(1 × 10 = 10 marks)